S2Q21

Question 21 - Stage 2

Let S be the set of positive integers that can be written in the form $2^a3^b5^c7^d$, where a, b, c, d are non-negative integers.

For example, all the integers from 1 to 10 are in S.

What (to 4s.f.) is the sum of the reciprocals of all the numbers in S? (Enter your answer in the usual way in the box below.)

Working

In [4]:

```
from itertools import product
from pyTools import roundSF

maxValue = 50 # If the answer is wrong, increase to be more accurate
s = set()

for a,b,c,d in product(range(maxValue),repeat=4):
    result = (2**a) * (3**b) * (5**c) * (7**d)
    s.add(result)

reciprocal = {1/i for i in s}

print(f"Total: {roundSF(sum(reciprocal),4)}")
```

```
Total: 4.375
```